

Deepak Mallampati

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Kent, OH - Remote-US/Europe; open to EU work authorization discussion

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) with a track record of mapping complex customer workflows to AI solutions, designing bespoke integrations, and deploying to production with measurable quality baselines. At Progress Solutions, worked with healthcare stakeholders to identify AI use cases, built LangChain integrations, and delivered full deployment lifecycle (~35% retrieval precision improvement, 90%+ grounded alignment, ~30% fewer post-deployment defects). Independent projects include Manga Lens (4 AI providers, 29-site integration architecture, shipped to Chrome Web Store end-to-end) and Dream Decoder (FastAPI + React multimodal orchestration). MS, Kent State University. Ready to bring integration design and deployment ownership to voice AI customer workflows.

CORE COMPETENCIES

Customer Workflow Mapping • Bespoke Integration Design • LLM & AI API Integration • Python & FastAPI • End-to-End Deployment Ownership • Technical Communication • RAG & Agentic Systems • LangChain • Full-Stack (TypeScript/React) • Rapid Prototyping

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee

USA - Healthcare Technology, AI Consulting, SaaS Healthcare

- Built **Retrieval-Augmented Generation (RAG)** systems for clinical knowledge retrieval and healthcare documentation search; implemented recursive semantic chunking and transformer-based embeddings; improved contextual retrieval precision by ~35% and reduced irrelevant retrieval by >30%. Retrieval-grounded response alignment exceeded 90% in evaluation.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction, care engagement scoring, and support prioritization; improved recall by 15-20% on high-risk categories while holding precision >90% via class weighting, stratified sampling, and threshold calibration.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by ~30%.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, support ticket logs; raised dataset reliability above 98% and cut downstream model instability by ~40%.
- Maintained HIPAA-conscious data governance: de-identification, data lineage documentation, evaluation audit trails, and stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas, Enterprise ERP, Industrial Automation

- Optimized SQL and **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform; reduced query execution time ~20% and data retrieval latency ~25%.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; reduced deployment errors >30% and improved release cycle time ~35-40%.
- Designed performance dashboards using **SQL Server DMVs and Grafana**; cut incident recurrence ~25% and improved root-cause resolution speed ~30%.
- Supported RBAC implementation and audit logging for financial modules in a compliance-sensitive environment.

Machine Learning Engineer

Hyderabad, India - Nonprofit Tech, Video Analytics, Applied NLP

- Built a **transformer-based video summarization system** across 5,000+ sessions; reduced manual review time by **60-70%** and achieved **~85% alignment** with human-curated highlights.
- Implemented **document Q&A with RAG**; reduced hallucinated outputs **~30%** and raised contextual accuracy **>85%**. Deployed as a Flask-based API for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity search for duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Python, LangChain, RAG, JSON Schema, Vector Search

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped · Chrome Web Store

Full-stack browser extension built independently - Manifest V3, content scripts, service workers. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/local). Multi-section capture pipeline, seven-day translation cache, per-domain selector configs for **29 manga/webtoon sites**, narrowed host permissions.

TypeScript, React, Manifest V3, multi-provider AI APIs, Chrome Web Store

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript app; coordinated multimodal APIs (Perplexity Sonar + Replicate image models). Introduced **intermediate structured prompt transformation layers** - improved contextual alignment **~30%** and first-pass image success rate **~25-30%**.

FastAPI, React, TypeScript, Vite, Tailwind, Perplexity API, Replicate

Driver Drowsiness Detection System - Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN pipeline with unified **YOLOv8** model; reduced inference latency **~30%**. Sliding-window confidence aggregation suppressed blink-driven false positives **~25%**; adaptive frame skipping and NMS tuning for stable real-time operation via OpenCV.

YOLOv8, OpenCV, Python, LabelImg, PyTorch

E-Farming Digital Marketplace founder · full-stack developer

Independently designed and shipped a PHP/MySQL/Bootstrap AgriTech marketplace connecting small-scale farmers with buyers. Onboarded **80-120 active users** in phase one; cart, reviews, and community blogging features.

PHP, MySQL, Bootstrap, full-stack marketplace

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

CERTIFICATIONS

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

LLM & GenAI: LLM application development, Retrieval-Augmented Generation (RAG), agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion, video analytics

Data & Analytics: Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation, NetworkX

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, cloud-ready deployment, observability/logging

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal Systems